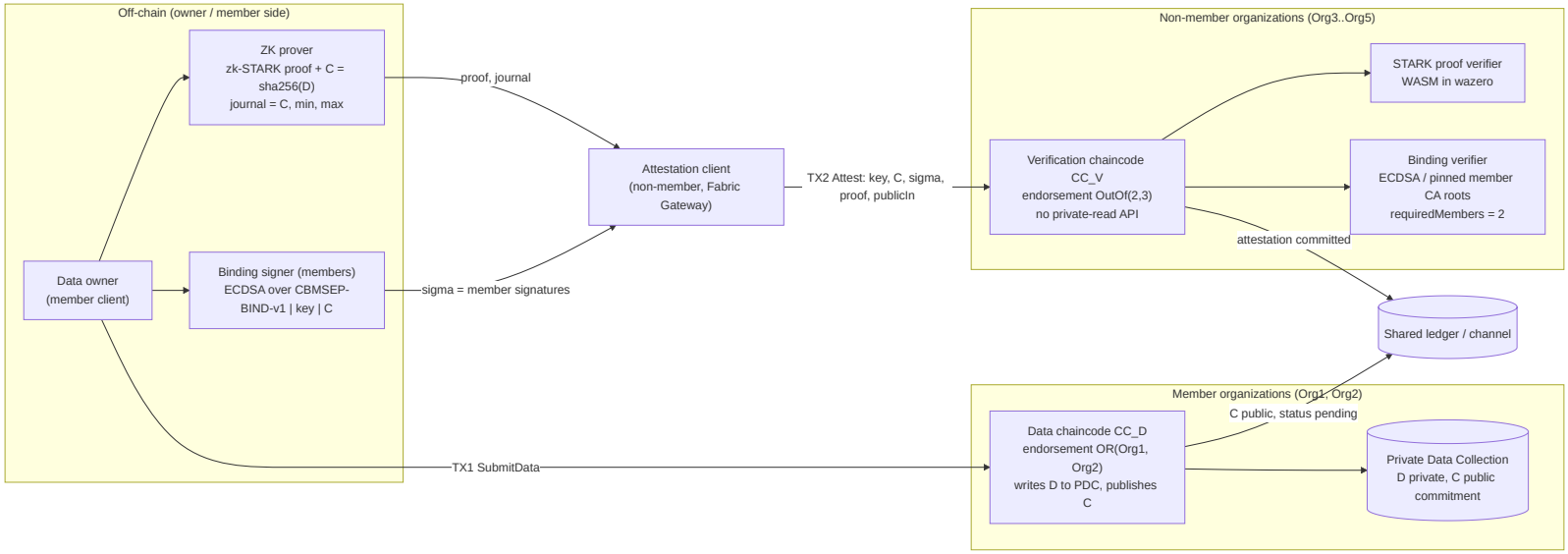
Off-chain (owner / member side)
ZK prover
zk-STARK proof + C =
sha256(D)
journal = C, min, max
Data owner
(member client)
Binding signer (members)
ECDSA over CBMSEP-
BIND-v1 | key | C
proof, journal
sigma = member signatures
Attestation client
(non-member, Fabric
Gateway)
TX2 Attest: key, C, sigma,
proof, publicIn
Non-member organizations (Org3..Org5)
Verification chaincode
CC_V
endorsement OutOf(2,3)
no private-read API
STARK proof verifier
WASM in wazero
Binding verifier
ECDSA / pinned member
CA roots
requiredMembers = 2
attestation committed
Shared ledger / channel
Member organizations (Org1, Org2)
C public, status pending
Data chaincode CC_D
endorsement OR(Org1,
Org2)
writes D to PDC, publishes
C
Private Data Collection
D private, C public
commitment
TX1 SubmitData